

Azure Virtual Machine Deployment & Web Hosting Assignment

Student Name	Jashika Sahu
Platform	Microsoft Azure
VM Name	PracticeVM
Resource Group	RG-Practice-VM
Public IP	20.244.84.44

Part 1 — Create Resource Group

1. Logged into Microsoft Azure Portal.
2. Opened Resource Groups section.
3. Created RG-Practice-VM resource group.
4. Selected Central India region.
5. Deployment completed successfully.

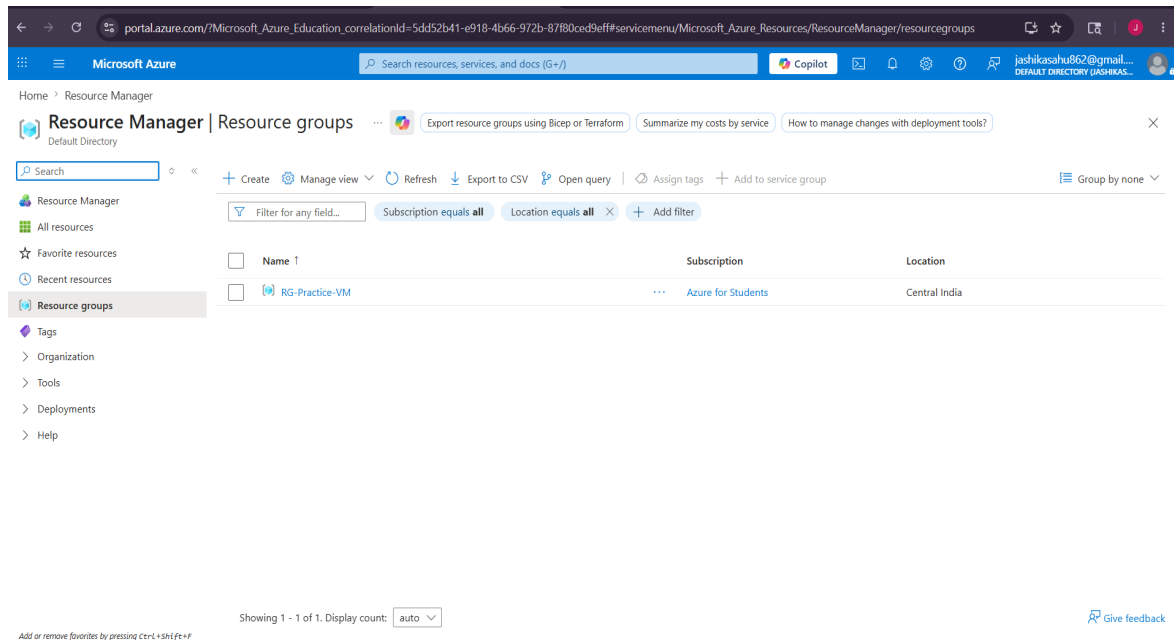


Figure 1: Resource Group created successfully.

Part 2 — Create Virtual Machine

1. Created Ubuntu Linux Virtual Machine.
2. Configured SSH authentication.
3. Enabled Public IP.
4. Allowed SSH and HTTP ports.
5. VM deployment completed successfully.

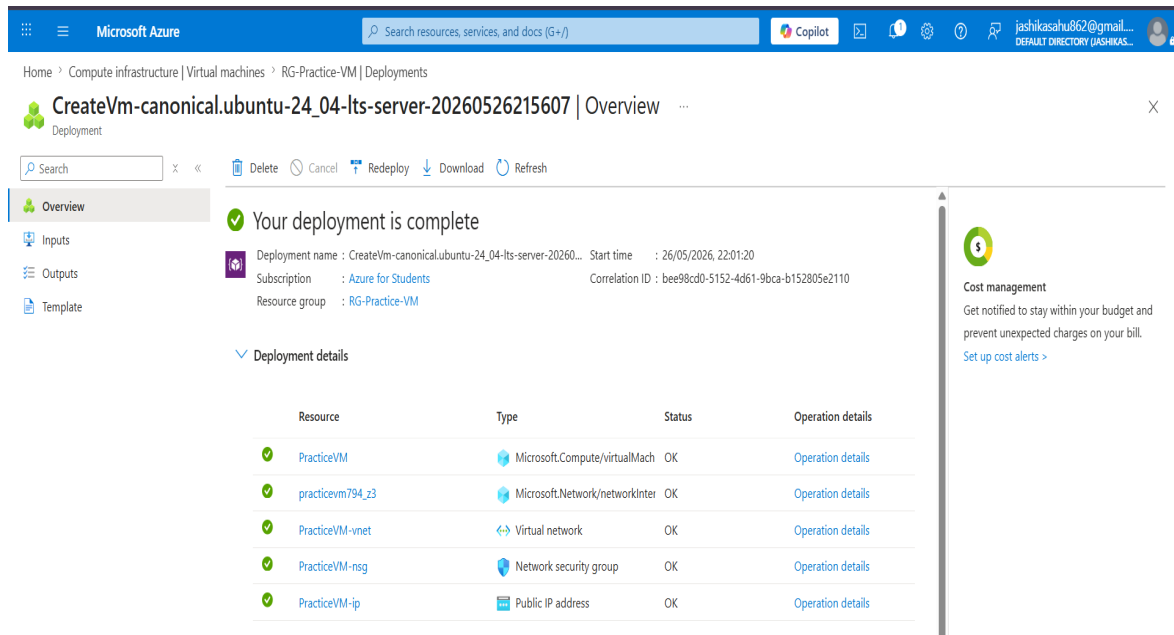


Figure 2: Azure Virtual Machine deployed successfully.

Part 3 — Connect to VM using SSH

1. Opened terminal in Windows.
2. Used SSH private key for authentication.
3. Connected to VM using Public IP.
4. Verified successful Ubuntu login.

```
C:\Users\nan\Downloads>ssh -i PracticeVM_key.pem azureuser@20.244.84.44
The authenticity of host '20.244.84.44 (20.244.84.44)' can't be established.
ED25519 key fingerprint is SHA256:V0uHlItKc37WZE5r6WwnAG03u25iF6z5Y4UP7s0xn0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.244.84.44' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1015-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue May 26 16:48:20 UTC 2026

System load:  0.08      Processes:    132
Usage of /:   5.8% of 28.02GB   Users logged in:  0
Memory usage: 33%      IPv4 address for eth0: 10.0.0.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@PracticeVM:~$ |
```

Figure 3: Successful SSH connection to VM.

Part 4 — Install Apache Web Server

1. Updated package repository.
2. Installed Apache2 server.

3. Started Apache service.
4. Enabled Apache service.

```
Enabling module authz_host.  
Enabling module authn_core.  
Enabling module auth_basic.  
Enabling module access_compat.  
Enabling module authn_file.  
Enabling module authz_user.  
Enabling module alias.  
Enabling module dir.  
Enabling module autoindex.  
Enabling module env.  
Enabling module mime.  
Enabling module negotiation.  
Enabling module setenvif.  
Enabling module filter.  
Enabling module deflate.  
Enabling module status.  
Enabling module reqtimeout.  
Enabling conf charset.  
Enabling conf localized-error-pages.  
Enabling conf other-vhosts-access-log.  
Enabling conf security.  
Enabling conf serve-cgi-bin.  
Enabling site 000-default.  
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.  
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.  
Processing triggers for ufw (0.36.2-6) ...  
Processing triggers for man-db (2.12.8-4build2) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
azureuser@PracticeVM:~$
```

Figure 4: Apache server installation completed.

Part 5 & 6 — Host Website and Access via Public IP

1. Created custom HTML webpage.
2. Hosted webpage inside Apache directory.
3. Opened website using Public IP address.
4. Verified successful website hosting.

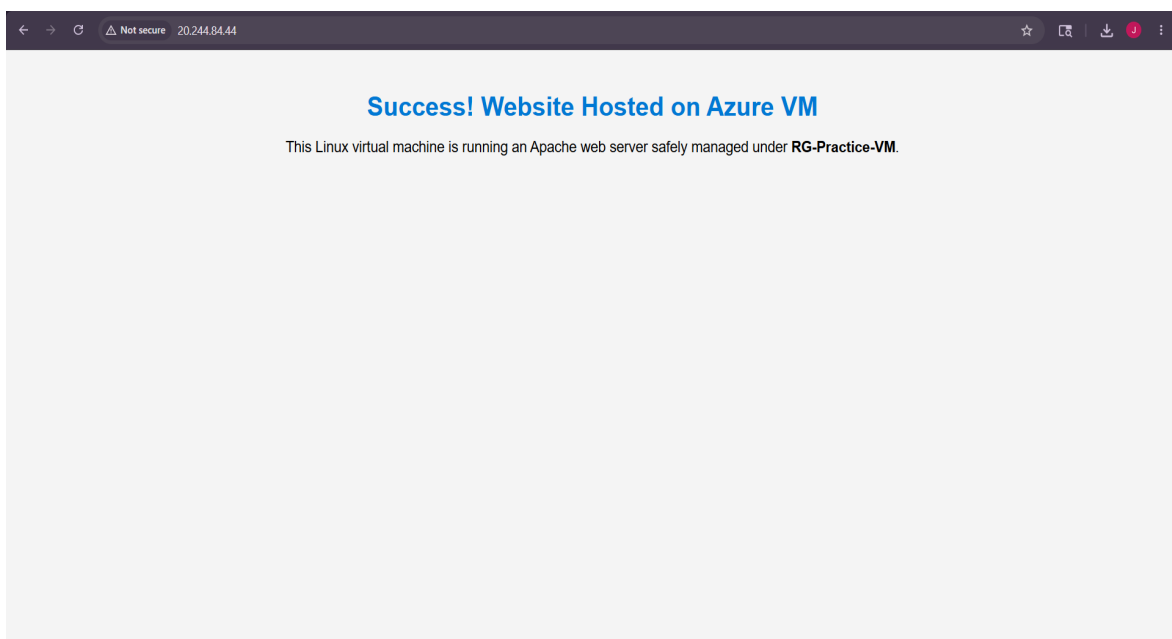


Figure 5: Website hosted successfully on Azure VM.

Commands Used

SSH Connection:

```
ssh -i PracticeVM_key.pem azureuser@20.244.84.44
```

Apache Installation:

```
sudo apt update
```

```
sudo apt install apache2 -y
```

```
sudo systemctl start apache2
```

```
sudo systemctl enable apache2
```

Conclusion

This assignment successfully demonstrated the deployment and configuration of a Linux Virtual Machine on Microsoft Azure. The Resource Group and Ubuntu VM were deployed successfully, SSH access was configured, Apache Web Server was installed, and a custom website was hosted successfully using the Public IP address.